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I. Earned Degrees

Ph.D.	2024	Georgia Institute of Technology	Industrial Engineering (Major); Machine Learning (Minor) Advisor: Prof. Jianjun Shi
M.S.	2024	Georgia Institute of Technology	Computational Science and Engineering
B.S.	2017	Xi'an Jiaotong University	Energy and Power Engineering

II. Employment History

University of Minnesota
Department of Industrial & Systems Engineering
Assistant Professor 08/2024–Present

Apple Inc.
Artificial Intelligence and Machine Learning Intern in AIML Organization 05/2021–08/2021

Georgia Institute of Technology
Research/Teaching Assistant/Graduate Instructor 08/2018–08/2024

III. Honors and Awards

A. International or National Awards

A.1. Best Paper Awards

- Best Student Paper Award, Physical and Engineering Sciences (SPES) and Quality and Productivity (Q&P), American Statistical Association (ASA) 2025
- Best Poster Award (Honorable Mention), Quality, Statistics & Reliability (QSR) Section of INFORMS 2024
- Best Track Paper Award (Winner), Quality Control and Reliability Engineering (QCRE), IISE 2023
- Best Student Paper Award (Winner), QSR Section of INFORMS 2022
- Best Track Paper Award (Finalist), Data Analytics & Information Systems, IISE 2022
- Best Student Paper Award Finalist, Data Analytics & Information Systems, IISE 2022
- Best Student Paper Award Finalist, Data Analytics & Information Systems, IISE 2021

A.2. Other Awards and Scholarships

- Inaugural Best Reviewer Award, IISE Transactions 2026
- Mary G. and Joseph Natrella Scholarship, ASA Q&P Section 2024
- John S.W. Fargher Jr. Scholarship, IISE 2023
- IISE Future Faculty Fellows (3F), IISE 2022
- Educational Foundation Scholarship, International Society of Automation (ISA) 2022
- China National Scholarship, Ministry of Education of P.R. China 2013–2015

B. Institute or School Awards

- MidCareer Fellow to the Institute for Pure and Applied Mathematics (IPAM) for its Long Program 'Multi-Fidelity Methods for Fusion Energy' 2026

- Sigma Xi Best Ph.D. Thesis Award, Georgia Institute of Technology 2025
- Angela P. & Reed J. Baker Research Excellence Award, School of ISyE, Georgia Tech 2023
- Outstanding Graduate Student Instructor of the Year, School of ISyE, Georgia Institute of Technology 2022
- Future Scholar Scholarship, Tsinghua University 2017

C. Awards won by Advisees

- Xinhan Yang: Robert E. Greiling, Jr. Scholarship for Graduate Studies in Renewable Energy, College of Science and Engineering, University of Minnesota 2026
- Deyi Kong: Best Student Paper Award Finalist, QSR Section of INFORMS 2025

IV. Grants and Contracts

Summary: $1.4M (personal share: $700K) in cumulative cash research funding secured as PI and lead proposal author from federal (NSF), internal (UMN Data Science Initiative), and industry (Samsung, Apple) sponsors, plus $660K (personal share: $540K) in in-kind high-performance computing resources secured as PI and Co-PI from NSF (NAIRR Pilot, ACCESS) and NVIDIA.

A. As Principal Investigator

[1]. **Title of Project:** Collaborative Research: Elements: Data-Agent: Agentic AI for Autonomous Data Quality Evaluation, Curation and Task-Aware Dataset Design for Vision-Based Industrial Inspection
Agency/Company: National Science Foundation (NSF), Office of Advanced Cyberinfrastructure (OAC), Cyberinfrastructure for Sustained Scientific Innovation (CSSI), NSF 22-632 (Award No. 2608818)
Total Dollar Amount: $600K; UMN Share: $399,967; Personal Share: $200,000
Role: PI (Lead PI of the collaborative award with Rensselaer Polytechnic Institute)
Period of contract: 08/2026–07/2029

[2]. **Title of Project:** Neural Refrigeration Cycle Network enabled Optimal Control of Closed-loop Fluid Infrastructure System
Agency/Company: Samsung Global Research Outreach (GRO) Award (one of 11 teams selected globally for this award in 2024)
Total Dollar Amount: $300,000
Role: Single PI
Period of contract: 02/2025–02/2027

[3]. **Title of Project:** Big VISION: Advancing Data Quality for Vision-Based Industrial Inspection
Agency/Company: UMN Data Science Initiative (internal seed grant)
Total Dollar Amount: $200,000 (no overhead)
Role: Single PI
Period of contract: 01/2025–12/2027

[4]. **Title of Project:** Automatic Data Quality Evaluation and Synthetic Data Generation
Agency/Company: NVIDIA Academic Grant Program Award (30k A100 GPU hours)
Total Dollar Amount: $61,800 (in-kind; at market rate of $2.06 per GPU hour)
Role: Single PI

Period of contract: 06/2025–12/2025

[5]. **Title of Project:** Differentiable HVAC System Simulator
Agency/Company: NVIDIA Academic Grant Program Award (2 RTX6000 Max-Q workstation GPUs)
Total Dollar Amount: $26,500 (in-kind; 2 GPUs at $13,250 NVIDIA list price)
Role: Single PI
Period of contract: 06/2026–

B. As Co-Principal Investigator

[1]. **Title of Project:** NAIRR Pilot: Data-Centric AI Framework for Defect Detection in Advanced Manufacturing: Automated Data Curation, Annotation, and Design (NAIRR250089)
Agency/Company: NAIRR Pilot
Total Dollar Amount: $238,800 (AWS service credits); Personal Share: $119,400
Role: Co-PI
Period of contract: 05/2025–05/2027

[2]. **Title of Project:** Differentiable Physics-Informed Digital Twin for Energy Optimization of Large-Scale HVAC Systems (ENG260010)
Agency/Company: NSF ACCESS, Discover (1,500,000 credits; Jetstream2 GPU)
Total Dollar Amount: $321,000 (in-kind; ACCESS estimate); Personal Share: $321,000
Role: Co-PI (PI: Ph.D. advisee Xinhan Yang)
Period of contract: 05/2026–05/2027

[3]. **Title of Project:** Surrogate-Based Adaptive Continuous Sensor Design via Contextual Bilevel Learning (MTH260138)
Agency/Company: NSF ACCESS, Discover (1,500,000 credits; NCSA Delta GPU)
Total Dollar Amount: $11,861 (in-kind; ACCESS estimate); Personal Share: $11,861
Role: Co-PI (PI: Ph.D. advisee Zhiping Li)
Period of contract: 09/2026–09/2027

C. As Proposal Author and Lead Researcher (Ph.D. Student)

[1]. **Title of Project:** Machine Learning Enabled Data Fusion for Anomaly Detection
Agency/Company: Apple Inc.
Total Dollar Amount: $300,000
Role: Proposal author and lead Ph.D. researcher (PI: Prof. Jianjun Shi)
Period of contract: 08/2022–08/2024
Status: Funded
Description: Conceived as an extension of my 2021 Apple summer internship, this project addressed challenges in process monitoring and quality inspection. I assisted Prof. Shi in writing the proposal and served as the Ph.D. student carrying out the funded work. Initially a one-year award, it was extended for a second year based on the strength of our results.

V. Research, Scholarship, and Creative Activities

Publications from work done at University of Minnesota are indicated by '*', and the student authors are indicated by underline. Google Scholar: Link

A. Refereed Publications and Submitted Articles

A.1. Published and Accepted Journal Articles

[J1]. Y. Wang, **S. Mou**, J. Shi, and C. Zhang, "DISCO: Disentangling Signal Restoration and Corruption Identification via Sample-Wise Latent Low-Rank Similarity*," *INFORMS Journal on Data Science*, 2026, Accepted.

[J2]. J. Liu, **S. Mou**, N. Gaw, and Y. Wang, "Uni-3DAD: GAN-Inversion Aided Universal 3D Anomaly Detection on Model-Free Products," *Expert Systems with Applications*, vol. 272, p. 126 665, 2025.

[J3]. **S. Mou**, R. Vemulapalli, S. Li, Y. Liu, C. Thomas, M. Cao, H. Bai, O. Tuzel, P. Huang, J. Shan, and J. Shi, "Synth4Seg-Learning Defect Data Synthesis for Defect Segmentation Using Bi-level Optimization*," *IEEE Transactions on Automation Science and Engineering*, 2025.

[J4]. **S. Mou** and S. Maguluri, "Heavy Traffic Queue Length Behaviour in a Switch Under Markovian Arrivals," *Advances in Applied Probability*, vol. 56, no. 3, pp. 1106–1152, 2024, Selected in Applied Probability Collection: October 2025 Queueing Models, as one of the ten articles which have been published in the Journal of Applied Probability or Advances in Applied Probability.

[J5]. Y. Wang, **S. Mou**, J. Shi, and C. Zhang, "Physics-Informed Neural ODE with Heterogeneous Control Inputs (PINOHI) for Quality Prediction of Composite Adhesive Joints," *IISE Transactions*, pp. 1–32, 2024, Best Student Paper Winner, ASA SPES+Q&P 2025.

[J6]. Z. Zhang, **S. Mou**, K. Paynabar, and J. Shi, "Tensor-Based Temporal Control for Partially Observed High-Dimensional Streaming Data," *Technometrics*, vol. 66, no. 2, pp. 227–239, 2024, Best Student Paper Winner, INFORMS QSR 2022.

[J7]. Z. Zhang, **S. Mou**, M. Reisi Gahrooei, M. Pacella, and J. Shi, "Federated Multiple Tensor-on-Tensor Regression (FedMTOT) for Multimodal Data Under Data-Sharing Constraints," *Technometrics*, vol. 66, no. 4, pp. 548–560, 2024.

[J8]. **S. Mou**, M. Cao, H. Bai, P. Huang, J. Shi, and J. Shan, "PAEDID: Patch Autoencoder Based Deep Image Decomposition," *IISE Transactions*, vol. 56, no. 9, pp. 917–931, 2023.

[J9]. Z. Chen, **S. Mou**, and S. Maguluri, "Stationary Behavior of Constant Stepsize SGD Type Algorithms: An Asymptotic Characterization," *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 6, no. 1, pp. 1–24, 2022.

[J10]. **S. Mou**, M. Biehler, X. Yue, J. Hunt, and J. Shi, "Sparse Sensor Placement Based Adaptive Control for High Precision Fuselage Assembly," *IISE Transactions*, vol. 55, no. 11, pp. 1133–1143, 2022, Best Student Paper Award Finalist, IISE DAIS 2022.

[J11]. **S. Mou** and J. Shi, "Compressed Smooth Sparse Decomposition," *INFORMS Journal on Data Science*, vol. 2, no. 1, pp. 60–80, 2022, Best Track Paper Finalist, IISE DAIS 2022.

[J12]. Z. Zhong, **S. Mou**, J. Hunt, and J. Shi, "Convex Relaxation for Optimal Fixture Layout Design," *IISE Transactions*, vol. 55, no. 7, pp. 746–754, 2022.

[J13]. Z. Zhong, **S. Mou**, J. Hunt, and J. Shi, "Finite Element Analysis Model-Based Cautious Automatic Optimal Shape Control for Fuselage Assembly," *Journal of Manufacturing Science and Engineering*, vol. 144, no. 8, p. 081 009, 2022.

[J14]. **S. Mou**, A. Wang, C. Zhang, and J. Shi, "Additive Tensor Decomposition Considering Structural Data Information," *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 4, pp. 2904–2917, 2021, Best Paper Finalist, IISE DAIS 2021.

[J15]. Z. Wang, **S. Mou**, Z. Wang, and Z. Feng, "Effect of Inlet Swirl Flow on the Migration of Hot Spots and End Wall Slot Leakage Flow," Chinese, *Journal of Engineering Thermophysics*, vol. 40, no. 2, pp. 275–281, 2019.

[J16]. **S. Mou**, Y. Luan, W. Ji, J. Zhang, and W. Tao, "An Example for the Effect of Round-Off Errors on Numerical Heat Transfer," *Numerical Heat Transfer, Part B: Fundamentals*, vol. 72, no. 1, pp. 21–32, 2017.

A.2. Conference Presentations with Proceedings (Refereed)

[C1]. X. Yang, L. Lu, and **S. Mou**, "Derivative-Informed Training of Neural Operators On-the-Fly via Sketched Tangent Consistency*," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.

[C2]. D. Kong, Z. Chen, S. Zhang, and **S. Mou**, "Natural Hypergradient Descent: Algorithm Design, Convergence Analysis and Parallel Implementation*," in *International Conference on Machine Learning (ICML), 2026*, Best Student Paper Award Finalist, QSR Section of INFORMS, 2026.

[C3]. **S. Mou**, X. Gu, M. Cao, H. Bai, P. Huang, J. Shan, and J. Shi, "RGI: Robust GAN-Inversion for Mask-Free Image Inpainting and Unsupervised Pixel-Wise Anomaly Detection," in *The Eleventh International Conference on Learning Representations (ICLR), 2023*, Best Track Paper Winner, IISE QCRE 2023, 2023.

[C4]. Z. Chen, **S. Mou**, and S. Maguluri, "Stationary Behavior of Constant Stepsize SGD Type Algorithms: An Asymptotic Characterization," in *ACM SIGMETRICS 2022*, Conference version of J8, Acceptance Rate 20%, 2022.

[C5]. **S. Mou**, M. Cao, Z. Hong, P. Huang, J. Shan, and J. Shi, "Synthetic Defect Generation for Display Front-of-Screen Quality Inspection: A Survey," in *SID Symposium Digest of Technical Papers*, Invited Paper, vol. 53, 2022, pp. 975–978.

[C6]. **S. Mou**, J. Chen, C. Zhang, and B. Wang, "A Data Fusion Framework for Fracture Toughness Modeling Using Multiple Sources of Data," in *International Symposium on Flexible Automation*, 2020, V001T07A010.

A.3. Book Chapter

[B1]. J. Shi, M. Biehler, and **S. Mou**, "Synergy of Engineering and Statistics: Multimodal Data Fusion for Quality Improvement," in *Multimodal and Tensor Data Analytics for Industrial Systems Improvement*. Springer, 2024, pp. 255–279.

A.4. Submitted Journal/Conference Articles

[SJ1]. D. Kong, S. Zhang, and **S. Mou**, *A New Approach to Bilevel Optimization: A Value-Centric Perspective via a Distributional Follower**, Under Review, ICLR 2027.

[SJ2]. X. Yang, M. Song, D. Kong, M. Kim, H. Um, and **S. Mou**, *Physics-Informed Neural Component Model Enabled Digital Twin of Large Scale HVAC Systems: Modularized Design, Differentiable Simulation, and Efficient Energy Optimization**, Under Revision, IISE Transactions, 2026.

A.5. Preprints

[D1]. H. Bai, **S. Mou**, T. Likhomanenko, R. Cinbis, O. Tuzel, P. Huang, J. Shan, J. Shi, and M. Cao, *VISION Datasets: A Benchmark for Vision-Based Industrial Inspection*, arXiv preprint arXiv:2306.07890, 2023.

B. Presentations

B.1. Invited Talks (Selected)

1. IISE QCRE Division Webinar Series, "Physics-Informed Learning Meets Large-Scale Optimization: A Two-Way Connection," Virtual, Nov 2026.

2. SIAM NY–NJ–PA Section Annual Meeting, "Derivative-Informed Training of Neural Operators On-the-Fly via Sketched Tangent Consistency," Rutgers University, NJ, Oct 2026.
3. Samsung Electronics, "AI/ML Modeling + Large-scale Optimization for Large-scale Complex Engineering Systems," Pyeongtaek, South Korea, Sep 2026.
4. International Workshop on Quality and Reliability Science & Technology (QRST), Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Tutorial, "Physics-Informed Machine Learning: Methodology and Applications," Beijing, China, Jun 2026.
5. International Workshop on Quality and Reliability Science & Technology (QRST), Academy of Mathematics and Systems Science, Chinese Academy of Sciences, "Large-Scale Bilevel Optimization and Its Application to Quality, Productivity, and Energy Efficiency Improvement in Complex Engineering Problems," Beijing, China, Jun 2026.
6. Quality and Productivity Research Conference (QPRC), "Uncertainty-Aware Hypergradient Descent for Bilevel Optimization," Santa Fe, NM, Jun 2026.
7. IPAM (UCLA), Workshop IV: Multi-Fidelity Methods to Enable Robust Optimization and Real-Time Control of Fusion Processes, "Derivative-Informed Training of Neural Operators on the Fly," Los Angeles, CA, May 2026.
8. University of Iowa, ISE Seminar, "AI/ML-Enabled Data Fusion for Complex Engineering Systems," Iowa City, IA, Feb 2026.
9. Honeywell X UMN Computer Vision Workshop, "AI/ML-enabled Data Fusion in Multistage Systems," Minneapolis, MN, Oct 2025.
10. Minnesota Supercomputing Institute (MSI), "AI/ML enabled quality and productivity improvement of advanced manufacturing systems–the convergence of engineering science, optimization, and statistics," Minneapolis, MN, Sep 2025.
11. IEEE CASE, "Finite Element Analysis Based Methods for Fuselage Shape Control," Los Angeles, Aug 2025.
12. Yonsei University, Graduate Seminar, Department of Industrial Engineering, "Robust Learning and Its Applications in Industrial Inspection: Nonlinear Settings Beyond Simple Statistical Prior Assumptions," Seoul, South Korea, May 2025.
13. Samsung Electronics, "AI/ML-Enabled Data Analysis for Quality Efficiency and Productivity Improvement in Complex Engineering Systems," Pyeongtaek, South Korea, Mar 2025.
14. Apple DMLI Weekly Sharing, "Physics-Informed Learning: Methodology and Applications in Engineering Design and Optimal Control," Virtual, Jan 2025.
15. MD&M Conference, "AI/ML-Enabled Quality and Productivity Improvement of Advanced Manufacturing Systems," Minneapolis, MN, Oct 2024.
16. CSE DSI ML Seminar (UMN), "AI/ML-Enabled Data Fusion for Complex Engineering Systems," Minneapolis, MN, Sep 2024.
17. Natrella Session of Joint Research Conference for Award Receipients, "Distributional-Assumption-Free Robust Learning for Fine-Grained Anomaly Detection," Jun 2024 (virtual).
18. Apple AIML APAC Talk, "RGI: Robust GAN-Inversion, Theory and Applications," Cupertino, CA, Oct 2022.

19. SID Display Week, "Synthetic Defect Generation for Display Front-of-Screen Quality Inspection: A Survey," San Jose, CA, May 2022.
20. Apple ML Summit, "DID: Deep Image Decomposition for Pixel-Level Anomaly Detection," Cupertino, CA, Feb 2022.
21. Georgia Tech ISyE Student Seminar, "Additive Tensor Decomposition Considering Structural Data Information," Dec 2021.
22. Various talks at INFORMS/IISE annual conferences

VI. Education

A. Courses Taught

A.1. University of Minnesota

Undergraduate Level

Semester, Year	Course Number	Course Title	Students
Spring 2025	IE 3522	Methods for Quality Engineering and Six Sigma	64
Fall 2025	IE 3521	Statistics, Quality, and Reliability	85
Spring 2026	IE 3522	Methods for Quality Engineering and Six Sigma	83

Graduate Level

Semester, Year	Course Number	Course Title	Students
Fall 2025	IE 8534	Physics-Informed Machine Learning: Methodology and Applications	16

Note on IE 8534: This interdisciplinary PhD course was newly developed by the instructor and bridges computational methods, large-scale optimization, and machine learning, while drawing applications from diverse fields. It has attracted students from a wide range of departments beyond Industrial and Systems Engineering, including Mechanical Engineering, Aerospace Engineering, Earth Sciences, Civil Engineering, Electrical Engineering, Chemical Physics, and Mathematics. Course website: Link

A.2. Georgia Institute of Technology

Undergraduate Level

Semester, Year	Course Number	Course Title	Role
Spring 2021	ISYE 3133	Engineering Optimization	Instructor
Fall 2021	ISYE 3133	Engineering Optimization	Instructor

Note on ISYE 3133: Served as the graduate student instructor for the studio sessions; received the Outstanding Graduate Student Instructor of the Year Award, School of ISyE.

B. Individual Student Guidance

B.1. Ph.D. Students (In Process)

[1]. Name: Deyi Kong
Advising Start Date: 2024

[2]. Name: Xinhan Yang
Advising Start Date: 2025

[3]. Name: Zewei Deng (Co-advised with Prof. Liyan Xie)
Advising Start Date: 2025

[4]. Name: Robert Weber
Advising Start Date: 2026

[5]. Name: Zhiping Li
Advising Start Date: 2026

B.2. M.S. Students

[1]. Name: Wan Wang
M.S. in: Computer Science
Advised: 2024–2025
Placement: Huawei Technologies

[2]. Name: Yu-Tong Chuang
M.S. in: Computer Science
Advised: 2025–2026

[3]. Name: Haoyu Wang
M.S. in: Industrial Engineering
Advised: 2025–2026
Placement: Contemporary Amperex Technology Co., Limited (CATL)

B.3. Undergraduate Students

[1]. Name: Michele Nguyen
B.S. in: Computer Science & Industrial Engineering (Undergraduate Research Award)
Advised: Spring 2025

[2]. Name: Gloria Le
B.S. in: Computer Science
Advising Start Date: 2025
Placement: Ph.D. student, Computer Science, University of Minnesota

B.4. High School Students

[1]. Name: Parth Rana
Wayzata High School
Advising Start Date: 2026

VII. Service

A. Professional Contributions

A.1. Society Officers, Activities, and Membership

- Editorial Board Memberships

1. Lead Guest Editor, *Machine Vision and Applications*, Special Issue for VISION'26 (Workshop on Vision-Based Industrial Inspection, CVPR 2026) 2026–present
2. Guest Editor, *Machine Vision and Applications*, Special Issue on Vision-Based Industrial Inspection, for VISION'25 (ICCV 2025) 2025–2026

- Leadership Roles:
  1. Student Board Member of IISE QCRE Section 2022–2024
  2. Organizer, Student Seminar Series, INFORMS QSR Section 2023–2024
- Award Committees:
  1. INFORMS QSR Best Refereed Paper Competition Committee 2025, 2026
- Membership:
  1. ASME: American Society of Mechanical Engineers
  2. ASA: American Statistical Association
  3. CVF: Computer Vision Foundation
  4. IISE: Institute of Industrial and Systems Engineers
  5. INFORMS: Institute for Operations Research and Management Science
  6. ISA: International Society of Automation

A.2. Organization and Chairmanship of Technical Sessions, Workshops and Conferences:

- Organization of Conferences and Workshops:
  1. Chair & Co-organizer, Workshop on Vision-Based Industrial Inspection, CVPR 2026 (Jun 2026, Denver, CO).
  2. Chair & Co-organizer, Workshop on Vision-Based Industrial Inspection, ICCV 2025 (Oct 19, Honolulu, Hawaii).
  3. Chair & Co-organizer, Workshop on Vision-Based Industrial Inspection, ECCV 2024 (Sep 30, Milano, Italy).
  4. Co-organizer, Workshop on Vision-Based Industrial Inspection, CVPR 2023 (Jun 19, Vancouver, Canada).

A.3. Reviewer

- NeurIPS/ICML/ICLR/AISTATS
- Management Science
- Technometrics
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Automation Science and Engineering
- IEEE Transactions on Emerging Topics in Computational Intelligence
- IISE Transactions
- INFORMS Journal on Data Science
- Journal of Intelligent Manufacturing
- Scientific Reports
- Journal of Computational and Graphical Statistics
- ASME Journal of Computing and Information Science in Engineering
- Machine Vision and Applications
- Computers & Industrial Engineering